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TinkerWithMe

Robot Arm & Claw · Premium Course

2 live sessions · 2 hrs each · 12–15 yrs · from KES 7,000

Premium course outline **PREMIUM**

Overview

Assemble and program a 3-DOF robot arm with a gripper claw and a joystick controller.

DETAIL	
Track	Arduino Robotics & Electronics
Format	2 live sessions · 2 hrs each
Delivery	Live online (1:1 or small group) or in person, you choose when booking
Ages	12–15 yrs
Level	Advanced
Price	from KES 7,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Needed at home for live online sessions; for in-person sessions we provide kits, so buy one only to keep building at home.
Parts pack	Optional, KES 2,500: 3-DOF robot arm frame + claw, 4 extra SG90 servos, 2 joystick modules
Standalone	No
Prerequisites	Servo motor control (free)

Programme

1. **Build & Wire.** Assemble a 3-DOF (degree-of-freedom) arm using SG90 servos and build a joystick controller with two analog sticks.

2. **Program & Challenge.** Calibrate movement ranges and add a gripper claw servo. Challenge: pick up and sort coloured objects by zone.

What they take home

🎁 Assembled robot arm

Why parents love it

“The same principles behind factory robots and surgical arms — in your child's hands.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Robot Arm & Claw**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.